DATA SOCIETY

Foundations of Data and AI Literacy for Managers

Participant Guide

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Class activities

Data-You use it more than you think

Time allotted: 5 minutes

Goal: To become more aware of the data or information you encounter daily and how it influences your decisions.

1. Consider the data you interact with regularly

- Take 5 minutes to think about the information you use regularly. This could include anything from news articles and social media posts to weather reports, maps, etc.
- Write down a list of 3-5 specific types of information/data that you rely on.

2. Identify characteristics of that data

- For each item on your list, analyze its characteristics:
  - What type of data is it?
  - Is it structured, semi-structured or unstructured?
  - What is the source of this information?

3. Reflect on your relationship with data

- How do you use this data to make decisions or support your choices?
- What data sources do you trust, and why do you consider them reliable?

Record your responses in the table below:

Data/ Information	Type and format of data	Source of data	Big/Small data	How do you use it? Why do you consider it reliable?

Data analytics case study

Time allotted: 15 minutes

Goal: To examine the scenario and classify the provided questions according to the four stages of analytics: descriptive, diagnostic, predictive, and prescriptive.

Scenario:

A prominent chain of fitness centers called Titan Fitness is experiencing an unfavorable trend due to its declining membership renewals and a rise in customer churn. Despite the chain's strong brand reputation, cutting-edge equipment, and wide range of classes, Titan Fitness is struggling to retain customers in an increasingly competitive market. Recognizing the urgency of this situation, the marketing team has decided to investigate the issue and develop a data-driven strategy to revitalize the business and restore customer loyalty.

The task force has access to a wealth of data from Titan Fitness's various systems:

- Customer membership data: Detailed records of all members, including demographics, membership type (monthly, annual, etc.), sign-up date, renewal history, and personal training sessions.
- Usage data: Records of gym visits, class attendance, and equipment usage collected through membership card swipes and booking systems.
- Customer feedback data: Surveys, online reviews, and customer service interactions capture member satisfaction, complaints, and suggestions.
- Financial data: Revenue and expenses related to membership fees, personal training, merchandise sales, and operational costs.
- Marketing data: Data on marketing campaigns across various channels (social media, email, local partnerships), including costs, reach, and conversion rates.

- Competitor data: Information on competing gyms and fitness studios, including their membership pricing, class offerings, and online reviews.
- Market trends data: Reports and analyses on fitness industry trends, emerging technologies, and changing customer preferences.

The following questions were addressed through the application of data analytics. For each question, select the data analytics stage that would be useful in finding a solution or gain the necessary insights.

Questions	**Data analytics stage**
1. Why are membership renewal rates declining?	Diagnostic Analytics
2. **Should we expand into new markets or demographics to reach a wider audience?**	Prescriptive Analytics
3. What strategies can be implemented to improve member retention?	Prescriptive Analytics
4. Which members are most likely to cancel their memberships in the next quarter?	Predictive Analytics
5. What is the average membership length before cancellation?	Descriptive Analytics
6. Are certain locations experiencing higher churn rates than others? If so, why?	Diagnostic Analytics
7. Which membership types (monthly, annual, etc.) have the highest and lowest renewal rates?	Descriptive Analytics
8. What are the most common reasons cited for membership cancellations in customer feedback?	Diagnostic Analytics
9. What will the projected membership churn rate be in the next year?	Predictive Analytics
10. Are there specific demographics or membership types with significantly higher churn rates?	Descriptive Analytics

Detect errors

Time allotted: 5 minutes

Goal: To critically examine a sample dataset and identify any gaps or inconsistencies that might affect its reliability or usefulness.

Take 5 minutes to evaluate the dataset and note down the inaccuracies in it.

	A	B	C	D	E	F
1	**Name**	**Last Name**	**DOB**	**Country**	**Country**	**Status**
2	Andrew	Jackson	11/4/87	Argentina	Buenos Aires	Citizen
3	Natalie	Walker	2/12/82	France		
4	Sam	Williams	3/30/80	Argentina	Mendoza	Permanent resident
5	Lilian	Perez	2-Jan-99	France	Nantes	Permanent resident
6	Sarah	Sanchez	1/8/78	Germany	Munich	Citizen
7	J	McAdam	7/18/00	England	London	Permanent resident
8	Oliver	Oliver	12/21/90	Ireland		Citizen
9	Natalie	Walker	2/12/82	France	Paris	
10	Iris	Nelson	10/20/95	India	Mumbai	Citizen
11	Natalie	Roberts	9/30/90	Germany	Munci	Citizen
12	Nina	Rose Turner	1/19/01	UK	London	Citizen
13	Tara	Morris	2/20/63	Delhi	India	Citizen
14	Paul		8/24/03	America	Buenos Aires	Permanent resident
15	PAULINE	BROOKS	7/7/98	India	Banglore	Permanent resident
16	Natalie	Walker	2/12/82	France	Paris	Citizen

Record your responses in the space below:

Data management challenges

Time allotted: 15 minutes

Goal: To identify and evaluate the core data management subject areas for the given scenarios.

With your small group, read and analyze the three scenarios describing data management challenges faced by a fictional company.

Then, complete the analysis worksheet for each scenario by identifying:

1. **Core data management issue(s)**
   What are the specific data-related problems in the scenario? (e.g., data quality, data security, data privacy, data storage)
2. **Evidence**
   What evidence from the scenario supports their identification of the issues?
3. **Impact**
   What are the potential consequences of these issues for the company and its stakeholders?

continue to the next page for the scenario #1

Scenario #1

An online retailer, serving millions of customers globally, has suffered a data breach due to a sophisticated phishing attack. Hackers gained unauthorized access to a database containing a treasure trove of sensitive customer information, including names, addresses, credit card details, and social security numbers.

Core data management issue(s)	
Evidence	
Impact	

continue to the next page for the scenario #2

Scenario #2

A multinational conglomerate, boasting a diverse portfolio of business units, is struggling with a data dilemma. Each department operates in silos and is responsible for storing data in their independent systems and formats.

Core data management issue(s)	
Evidence	
Impact	

continue to the next page for the scenario #3

Scenario #3

A marketing team is trying to launch a targeted campaign but cannot find the customer demographics data. They discover multiple outdated spreadsheets scattered across various departments, making it impossible to get a clear picture for their campaign. Meanwhile, the data science team struggles to incorporate valuable but poorly documented social media sentiment data, hindering their ability to provide accurate insights.

Core data management issue(s)	
Evidence	
Impact	

Data governance maturity assessment

Time allotted: 10 minutes

Goal: To understand your organization's data maturity level based on three key areas: People, Processes and Technology.

Read each statement in the assessment below carefully and select the answer that best reflects your organization's current state.

Assign points based on your response:

1 point: Primitive

2 points: Progressing

3 points: Scaling

4 points: Optimized

Calculate your score for each section to determine your overall maturity level.

Section 1: PEOPLE

Component	Primitive (1)	Progressing (2)	Scaling (3)	Optimized (4)	Your Score
Data roles and responsibilities	No dedicated data governance roles. Data experts exist but without formal titles.	Data governance leader recruited. Data owners and stewards defined for key domains.	Clear data roles and responsibilities defined. Data Managers recruited.	All data roles & responsibilities are assigned and well-defined.	1
Data governance committees	No clear instances to discuss data governance issues.	First Data governance council established.	Data governance committees established with different roles & sponsors.	Vivid data community based on data ownership. Efficient operating model with IT.	1

Data domains	Few data domains and families are informally defined and partly managed.	Some data domains are defined and managed with central support.	Key data domains are defined and managed.	All data domains are clearly defined and managed.	1
Data ownership	No clear data ownership established.	Data owners identified for key data domains.	Data ownership embedded in data governance framework.	Data ownership is a core principle, driving accountability and collaboration.	2

Section 1 total score - 5

Section 2: PROCESSES

Component	Primitive (1)	Progressing (2)	Scaling (3)	Optimized (4)	Your score
Data modelling	Limited or no data modeling, i.e., data is not well organized & there are no clear structures for how data is stored.	Basic data models for critical data domains.	Data models used for most data domains.	Comprehensive data models used across the organization, ensuring easily accessible data.	4
Data process documentation	Low level of documentation.	Documentation of some data processes.	Standardized and centralized data process documentation across domains.	Mature and documented data processes across all data domains.	4
Data process optimization	Data processes are partly ineffective and have high potential for optimization.	Team efforts to optimize data processes.	Focus on transversal efforts to optimize data processes.	Ability to scale efficiency to support strategic projects and AI initiatives.	4
Data standards	No defined data standards.	Documentation of standards for critical data elements.	Standards documented for most domains.	Standards documented for all domains.	2

Section 2 total score - 14

Section 3: TECHNOLOGY

Component	Primitive (1)	Progressing (2)	Scaling (3)	Optimized (4)	Your score
Data catalog	Emergent data governance tools through Excel files.	First centralized issue log for data quality.	Data catalog and business glossary are implemented to standardize terminology. Key domains have a centralized issue log for tracking and resolving data-related issues.	Data catalog and business glossary are widely used and integrated across key domains, ensuring consistency and accessibility of data definition.	4
Data mapping	No structured and centralized visibility over data/system landscape.	Mapping of critical data sources and systems, interdependencies & flows.	Identification of all data elements through clear mapping.	Effective data structuration. Mapping tool well adopted and frequently used.	4
Data quality control	Low level of data quality.	Initiatives to control data quality & accessibility needed to be scaled to improve operational efficiency.	Data quality processes implemented for key data domains.	Data Quality tools fully implemented.	4
Self-service analytics	Self-service analytics are difficult.	Basic self-service analytics tools for key data domains.	More data domains with self-service analytics.	All data domains accessible by self-service.	4

Section 3 total score - 16

Your total maturity level is - 35

Guide to Maturity Level Ranges:

12 - 21: Primitive

22 - 30: Progressing

31 - 39: Scaling

40 - 48: Optimized

Field trip

Time allotted: 5 minutes

Goal: To discover the potential of artificial intelligence in facilitating efficient text summarization.

Open your web browser and go to *http://Amazon.com*. Search for a product you're interested in (e.g., a book, electronic device, household item) and click on a product that has a decent number of reviews.

Look for the "Customers Say" section which is usually located near the customer reviews. Take a few minutes to carefully read the summarized insights provided in the "Customers Say" section.

Next, reflect on your experience and jot down your thoughts based on the following questions:

- How does this generated summary impact your experience as a potential Amazon customer?

- How might you benefit from a similar text summarization tool in your own work?

AI brainstorm

Time allotted: 10 minutes

Goal: To reflect on AI and its components and examine potential applications and implications in your work/field.

Take 10 minutes to reflect on the applications and implications of AI and ML methods for a data project and log your thoughts based on the questions below. When logging thoughts, think about your own workplace or industry.

Question	Response / Thoughts / Notes
What AI concept did you find most interesting today, and why?	
What are some of the biggest challenges or pain points you face in your daily work? *(e.g., repetitive tasks, data analysis, information overload, creative blocks)*	

How could AI tools and technologies potentially address these challenges? *(e.g., automation, pattern identification, personalized recommendations, content generation)*	
Imagine a specific AI tool designed for your job. What would it do? *(e.g., an AI assistant that prioritizes emails, a tool that generates reports from raw data, an AI that helps design presentations)*	

Data tasks and tools reflections

Time allotted: 10 minutes

Goal: To encourage critical thinking about the diverse roles, skills, and technologies contributing to your organization.

Take 10 minutes to carefully consider the different types of personnel and tools that your organization offers. Use the questions given below to structure your reflection.

Question	**Response / Thoughts / Notes**
Who do you usually work with on projects or tasks?	
How do you communicate and collaborate with others in your department? Is there a collaboration tool that you use?	

How effectively do different teams in your organization collaborate on data projects?	
What tools/technologies do you use most often in your work?	
Do you find these tools easy to use? Why or why not?	

Data and AI goals

Time allotted: 10 minutes

Goal: To consolidate learning from the course and set goals for continued growth and development.

Think back on what you learned and write down your thoughts for the following:

Short-term goals (next 3 months)	
Mid-term goals (6-12 months)	
Long-term goals (beyond 1 year)	

Additional resources

Data and AI glossary

Algorithm – An algorithm is a series of steps to accomplish a task (a set of directions that gets you from point A to point B, if you will). It can be a thought exercise, a series of mathematical expressions, or a piece of code (or pseudo-code). Algorithms are used in everyday life, and many industries. They are an essential building block of data science, analytics, and any other quantitative field.

API – An application programming interface (API) is an "entryway" to a computer system (such as a database) that allows you to access, retrieve, and edit its components. Most often APIs are used as data-communication mediums between applications. They are an essential part of scalable data-centric applications, research projects that are built around data, and anything else that requires automated data access.

Artificial Intelligence – Artificial intelligence (AI) is the apparent ability of machines to act "intelligently."

Bayes Theorem – Bayes' theorem is used to calculate conditional probability of an event given the knowledge of conditions that might be related to the event. Conditional probability is the probability of an event occurring given another related event has already occurred.

Example: We want to know the probability of A(ge), given the diagnosis of C(ancer). This quantity is unknown to us and hard to estimate. We could use Bayes Theorem to do that, because we have the probability of C(ancer) given the A(ge), the probability of A(ge), and the probability of C(ancer) like so: P(Age given Cancer) = P(Cancer given Age) *P(Age)P(Cancer)

Big Data – Big Data is a term that describes a large volume of data—both structured and unstructured. It is produced in such quantities that it cannot be ingested or processed by a single machine at once (even a large machine like a supercomputer), so special tools and techniques are needed to process and store it.

Note: This term is often misused; most data that is called Big Data is not really that big. True Big Data is produced by things like Google Searches, satellite *imagery of the Earth, climate simulations used by weather services, sensor data generated by cell towers, etc.*

Classification – Classification is a supervised machine learning method where the output variable is a category, such as "yes" or "no."

Clustering – Clustering is an unsupervised machine learning method used to discover groupings that are inherent in the data.

Clickstream – A clickstream is a record of a user's activity on the Internet, including every Web site that the user visits, how long the user was on a page or site, in what order the pages were visited, and newsgroups that the user participates in and even the e-mail addresses of mail that the user sends and receives.

Data governance – The management of the overall quality, integrity, relevance, and security of available data.

Data Lake – A data lake is a system or repository of data stored in its natural/raw format, usually object blobs or files.

Note: Data lakes and data warehouses are both widely used for storing big data, but they are not interchangeable terms. A data lake is a vast pool of raw data, the purpose for which is not yet defined. A data warehouse is a repository for structured, filtered data that has already been processed for a specific purpose. (Source: https://www.talend.com/resources/data-lake-vs-data-warehouse/)

Data Mining – Data mining is a study of extracting information from structured/unstructured data taken from various sources.

Data Science – Data science is an interdisciplinary field that combines elements of mathematics, statistics, logical thinking, programming, and a range of domain knowledge from various fields, which is used to solve day-to-day problems using a combination of methods and tools from the above disciplines.

Data Warehouse – A data warehouse is electronic storage of a large amount of data, which is designed for query and analysis. It is typically used to connect and analyze data from heterogeneous sources.

Note: Data lakes and data warehouses are both widely used for storing big data, but they are not interchangeable terms. A data lake is a vast pool of raw data, the purpose for which is not yet defined. A data warehouse is a repository for structured, filtered data that has already been processed for a specific purpose.
(Source: https://www.talend.com/resources/data-lake-vs-data-warehouse/)

Data Visualization – Any attempt to make data more easily digestible by rendering it in a visual context (e.g., charting, graphing, etc.).

Database – A database is a structured collection of data. The collected information is organized in a way such that it is easily accessible by the computer. Databases are built and managed by using database programming languages.

Dataset – A dataset is a collection of data, which is organized into some type of data structure. Several characteristics define a dataset’s structure and properties. These include the number and types of the attributes or variables, and various statistical measures applicable to them, such as standard deviation.

Deep Learning – Deep learning is a branch of machine learning that uses deep artificial neural networks. Artificial neural networks are types of machine learning algorithms that are built based on the idea of how a brain works with its neurons connected to each other and transmitting signals. Artificial neural networks are usually built in layers, those that have 3+ layers are considered “deep” and fall into the deep learning bucket.

Graph Analysis – Also known as **network analysis**. Graph analysis is an analysis of structures that model pairwise relationships between objects. The objects in graph analysis are called nodes. Their relationships (a.k.a. connections) are called edges. Graph analysis can be used across many industries, but the most common use case is analysis of social networks.

Machine Learning – Machine learning is the computational process wherein a machine “learns” and adjusts its behaviors based on feedback from data. Usually manifesting as an adaptable algorithm, machine learning helps computers predict outcomes without explicit human input.

Model – A model is a simplified replica of any real-life phenomenon / object at smaller scale. In quantitative disciplines, a model is usually a mathematical description of such a phenomenon / object.

Example: An example is a model to predict the level of education based on age. In a simplified way it looks like this:

Level of education = some quantity + some quantity * age (a linear model)
Although in real life there are many more factors and variables, this model could potentially display a general trend.

Open Data – Open data is data that can be freely used, shared and built-on by anyone, anywhere, for any purpose.

Outlier – An outlier is an observation that appears far away and diverges from an overall pattern in a sample.

Regression – Regression is a supervised learning method where the output variable is a real value, such as "amount" or "weight" and the input variable(s) is a real value, such as "size" or "height." The output variable depends on the input variable(s). The relationship between the input and the output are usually recorded in a mathematical model.

Structured Data – Structured data is data that has been organized into a formatted repository, typically a database, so that its elements can be made addressable for more effective processing and analysis.

Supervised Learning – Supervised learning is a type of learning in which we teach or train the machine using data that is well labeled, meaning some data is already tagged with the correct answer class (or group to which it belongs).

Text Mining – Text mining is the process of converting unstructured text data into meaningful and actionable information.

Unstructured Data – Unstructured data is information that either does not have a pre-defined data model or is not organized in a pre-defined manner.

Unsupervised Learning – Unsupervised learning is the training of machines using information that is neither classified nor labeled and allowing the algorithm to act on that information without guidance.

Popular analysis tools

	R	Python	Excel
Learning curve	Steeper learning curve for people without a programming background	Can be easy to learn for people without a programming background	Easy to learn for any analyst
Automation	Once you write commands you won't have to re-do the work, just upload a new data set	Once you write commands you won't have to re-do the work, just upload a new data set	New data sets are not always plug and play with your analysis
Analyzing data	Lots of libraries contributed by a broad user community	Over 6,500 packages contributed by the community including top academics	Limited to any particular version
Speed	In-memory, only limited by your computer's RAM	In-memory, only limited by your computer's RAM	Can be slower unless an enterprise configuration is used
Type of data	Reads data of almost any type	Reads data of almost any type	Limited to xlsx and csv files unless macros are used

	R	Python	Excel
Compatibility	Compatible with almost any data output, storage or processing platform	Not as compatible as Python with some data architecture systems, may need custom-built interfaces	While macros enhance compatibility, Excel is comparatively limited
Data manipulation	Very flexible data manipulation	Very flexible data manipulation, augmented by numerous data processing and manipulation packages	Color-coded formulas can be easier to use but have a more limited functionality
Seeing data	Command line presentation unless visualization libraries are used	Spreadsheet-like view function that can be less intuitive to navigate	Easy to navigate spreadsheet
Graphics	Cutting edge graphics, however advanced coding and JavaScript knowledge may be necessary	Cutting edge graphics including dynamic visualizations, maps, network graphs, etc.	More limited options based on pre-set drop-down menus (unless macros are used)
Cost & platform	Free, any platform	Free, any platform	Hundreds of dollars, functionality on a Mac does not always mimic a PC

Additional reading & reference

Doing Data Science *by Cathy O'Neil & Rachel Schutt*

Data Science for Business *by Foster Provost & Tom Fawcett*

Data Smart *by John W. Foreman*

Mining the Social Web *by Matthew A. Russell*

Predictive Analytics *by Eric Siegel*

Analyzing the Analyzers *by Harlan Harris, Sean Murphy and Marck Vaisman*

Use cases and additional examples

BNY Mellon advances artificial intelligence tech across operations

Combining Satellite Imagery and Machine Learning to Predict Poverty

New York City uses "nudges" to reduce missed court dates

Proof of concept: Using predictive modeling to prioritize building inspections

Recruiting Chatbots in 2021: In-Depth Guide

Tactical Institute: Protecting people and communities with pre-emptive social media threat analytics

What Wal-Mart Knows About Customers' Habits